MGSC 706: Management Research Statistics

Lecture Notes

PhD Core Course in Management Research Statistics

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These notes cover the course lecture material for the PhD core course in management research statistics.

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MGSC 706 is a core doctoral course in management research statistics. Its scope proceeds from probability to sampling, statistical inference, hypothesis testing, regression, and model building. Although formulas are essential, the course emphasizes business applications and interpretation. Our plan is to demonstrate Python in a cloud environment, while allowing students to use other statistical platforms such as R or SPSS when appropriate.

The pedagogical principle is learning by doing. Manual calculations are useful for understanding definitions, but real research datasets require software. The deeper goal is to learn how a statistical quantity maps to a research question, how to select the correct comparison group, and how to interpret output rather than merely generate it.

Assessment note. There will be four assignments that are worth 10% each, the midterm exam is worth 20%, the final exam is worth 30%, and participation is worth 10%.

1 | Lecture 1: Probability

Sample spaces • Conditional probability • Simpson's paradox • Independence • Covariance • Bayes' rule

1.1 Executive summary

Probability is the language that connects raw data to statistical reasoning, and this lecture builds that language from the ground up. The methodological message is as important as any formula: begin with transparent, model-free summaries of the data; use those summaries to identify patterns and candidate mechanisms; and only then turn to formal inference to judge whether the pattern is stable or merely an artifact of sampling variation. Mathematical machinery is not a substitute for research intuition, and intuition is not a substitute for rigorous analysis — strong empirical work moves back and forth between the two.

Four technical pillars organize the lecture. First, *empirical probability* is estimated by relative frequency in observed data. Second, *conditional probability* restricts attention to a subgroup and therefore changes the denominator used in the calculation. Third, *independence and covariance* describe whether two events co-occur more or less often than their marginal probabilities alone would predict. Fourth, *Bayes' rule* reverses the direction of conditioning, turning a prior belief into a posterior belief once new evidence arrives.

Throughout, a single applied case — graduate admissions by gender and department — carries the running example, while a set of smaller vignettes (a fair die, masks and COVID, online ad clicks, and defective parts from two suppliers) isolate individual ideas. A closing theme, echoed in several places, is that probability is also the mathematical core of modern generative AI: a language model is, at every step, estimating a conditional probability distribution over its next token.

1.1.1 Learning objectives

1. Distinguish a sample space, an event, a subgroup, and a random variable, and recognize different variable types.
2. Compute empirical, joint, marginal, and conditional probabilities from counts, including under counting rules for uniform outcomes.
3. Explain why the conditioning event determines the denominator in a conditional probability.
4. Decompose an aggregate rate into subgroup-specific rates and subgroup weights.

5. Diagnose Simpson's paradox and separate within-group comparisons from compositional effects.
6. State the chain rule, the multiplication criterion for independence, and the covariance of binary indicators.
7. Apply Bayes' rule and the law of total probability to update a prior probability after observing evidence.
8. Separate descriptive association, statistical significance, and causal interpretation.
9. Connect conditional probability, independence, and the chain rule to how generative AI models — especially language models — represent and update uncertainty.

A practical research workflow runs through the lecture's examples: inspect the raw records and understand what each variable represents; compute simple counts, proportions, and subgroup comparisons before fitting any model; form a provisional explanation for the observed pattern; use standard errors, confidence intervals, hypothesis tests, or regression to assess uncertainty and competing explanations; return to the data when formal analysis contradicts the initial intuition, since research is iterative rather than strictly linear; and report contradictory evidence and unsuccessful replications rather than selecting only the result that supports the preferred story.

1.2 Probability Models, Sample Spaces, and Events

A probabilistic model is a mathematical description of an uncertain situation. It must be in accordance with a fundamental framework that we discuss in this section. Its two main ingredients are listed below and are visualized in Fig. 1.1.

Elements of a Probabilistic Model

- The **sample space** Ω, which is the set of all possible outcomes of an experiment.
- The **probability law**, which assigns to a set A of possible outcomes (also called an **event**) a nonnegative number $P(A)$ (called the **probability** of A) that encodes our knowledge or belief about the collective "likelihood" of the elements of A. The probability law must satisfy certain properties to be introduced shortly.

1.2.1 Defining a probability

A probability is a normalized number measuring how likely an event is. A probability of zero means impossibility within the model, a probability of one means certainty, and intermediate values quantify uncertainty:

$$0 \leq P(A) \leq 1.$$

1.2.2 Sample spaces, events, and variable types

A probability model has three ingredients: an experiment, a sample space, and probabilities assigned to outcomes. The sample space, written Ω, contains every elementary

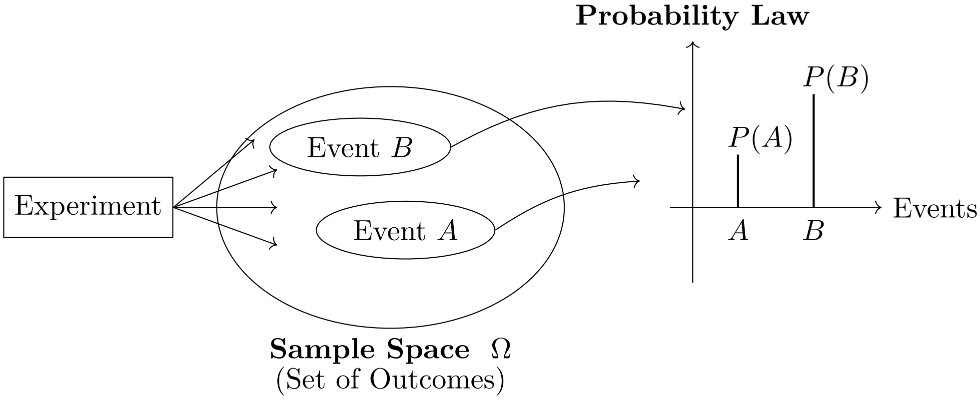


Figure 1.1: The main ingredients of a probabilistic model.

outcome the model recognizes. An event A is any subset of Ω and may contain one or many elementary outcomes:

$$A \subseteq \Omega, \qquad \sum_{\omega \in \Omega} P(\omega) = 1.$$

This normalization axiom means that increasing the probability of one outcome requires decreasing probability elsewhere — the same constraint enforced by a softmax layer in a neural network, discussed below.

Before assigning probabilities, it helps to classify the variables involved. *Nominal categorical* variables (gender, department, admission decision) carry distinct labels with no natural order. *Ordinal* variables (a Rensis Likert rating, class rank) have ordered levels whose distances need not be equal. *Quantitative* variables (GPA, age, sales, waiting time) carry numerical magnitude, so arithmetic on them is meaningful, subject to measurement assumptions. Table 1.1 summarizes the distinction and the summaries each type supports.

Table 1.1: Measurement types and their mathematical implications

Type	**Defining feature**	**Examples**	**Typical summaries**
Nominal categorical	Distinct labels; no natural order	Gender; department; admission decision	Counts, proportions, mode
Ordinal	Ordered levels; distances need not be equal	Rensis Likert rating; class rank	Median, quantiles, rank methods
Quantitative	Numerical magnitude; arithmetic is meaningful	GPA, age, sales, waiting time	Mean, variance, regression

1.2.3 Elementary outcomes versus events

An elementary outcome is indivisible at the model's chosen resolution; an event is any collection of elementary outcomes. The same real-world description can generate different sample spaces depending on the question asked. If the experiment is "draw one marble and record its color," the outcomes are red, green, and yellow. If the question is instead "how many red marbles appear in two draws," the outcomes are 0, 1, and 2, even though the same underlying ordered color pairs are being observed at finer resolution.

1.3 Frequentist Approach: Probability as relative frequency

When outcomes are observed frequently, probability is estimated by relative frequency of the outcome. Let Ω be the set of all units under consideration and let $E \subseteq \Omega$ be an event with $n(E)$ satisfying records:

$$\widehat{P}(E) = \frac{n(E)}{N}. \tag{1.1}$$

An empirical probability lies in $[0, 1]$; multiplying by 100 expresses it as a percentage. In repeated random samples of size m, the expected number of observations satisfying E is approximately $mP(E)$ — for $m = 100$ and $P(E) = 0.459$, the expected count is 45.9, or about 46.

The running case for this lecture is a graduate-admissions dataset of $N = 2{,}844$ applicant records, each carrying an applicant identifier, gender, department, and admission outcome. Of these applicants, 1,306 were admitted, so the overall empirical acceptance probability is

$$\widehat{P}(A) = \frac{1306}{2844} = 0.459 = 45.9\%. \tag{1.2}$$

This number supports three compatible readings: 45.9% of the observed applicants were accepted; a randomly selected record from this dataset has empirical acceptance probability 0.459; and a random sample of 100 comparable applicants would contain about 46 accepted applicants on average, an interpretation that depends on random sampling from the relevant population.

1.3.1 Counting with and without replacement

Counting rules supply probabilities only once a symmetry assumption is justified. If all elementary outcomes are equally likely,

$$P(A) = \frac{|A|}{|\Omega|} \qquad \text{(under uniformity)}. \tag{1.3}$$

For two sequential draws from three colors *with replacement*, the ordered sample space has $3 \times 3 = 9$ outcomes, since each position independently has three possibilities; if all colors and draws are symmetric, each outcome has probability $1/9$. *Without replacement*, and with one marble of each color available, only $3 \times 2 = 6$ ordered pairs remain. The lesson is that a counting formula should never be applied before the experimental protocol — inventory, replacement, and whether order matters — has been specified; most probability errors begin with an ambiguous sample space rather than difficult arithmetic.

1.3.2 From uniform assumptions to more complicated models

Uniform distributions are a useful starting model, but observed systems are rarely uniform: a biased die, unequal marble counts, language frequencies, and image structure all produce unequal probabilities. Section 1.3's empirical estimator, $\widehat{P}(A) = n(A)/N$, replaces the uniformity assumption with observed relative frequency, and the estimate can be refined as more data accumulate. A generative model performs a far more structured version of this same task, sharing statistical strength across contexts rather than independently counting every possible sentence or image.

1.3.3 Large language models as a motivating example

A large language model is, at its mathematical core, a probability model built on exactly the three ingredients above.

Sample space and elementary outcomes. Fix a vocabulary $\mathcal{V} = \{v_1, \ldots, v_K\}$ of tokens (roughly, word pieces). The elementary outcomes of the model's basic random experiment are the individual tokens in $\mathcal{V}$; choosing the next token is a draw from this space, just as choosing a face is a draw from $\{1, \ldots, 6\}$ for a die.

Events. A sentence, or more generally any token sequence $w_1 w_2 \cdots w_n$, is an *event*: a specific ordered combination of elementary outcomes, analogous to a specific hand of cards or a specific pair of dice values. Longer passages are simply more restrictive events, built by intersecting next-token outcomes across positions — the sequence-level probability is a composite of elementary draws, not a new primitive.

Probability law. What makes a language model a probability model is how it assigns probabilities to these outcomes and events. At each position, a deep neural network maps the preceding context to a vector of real-valued *scores* $z_1, \ldots, z_K$, one per vocabulary token; these scores by themselves are not probabilities (they need not be positive or sum to one). Softmax converts them into one:

$$P(\text{token} = v_i \mid \text{context}) = \text{softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}. \tag{1.4}$$

Every output is positive and the outputs sum to one, so (1.4) always defines a valid probability law over the elementary outcomes $\mathcal{V}$, satisfying the normalization axiom regardless of what the underlying network computes. This distribution is explicitly *conditional*: $P(\text{school})$ as a bare marginal ignores context, whereas the model estimates $P(\text{next token} = \text{“school”} \mid \text{context})$, and every additional prompt token changes the conditioning information and shifts this distribution — search autocomplete is a simpler, everyday version of the same idea.

The probability of an event — a whole sentence $w_1 \cdots w_n$ — follows by chaining these conditional elementary-outcome probabilities via the multiplication rule:

$$P(w_1, \ldots, w_n) = \prod_{t=1}^{n} P(w_t \mid w_1, \ldots, w_{t-1}),$$

each factor computed by the same score-then-softmax mechanism at position t. Generating text amounts to repeatedly sampling elementary outcomes from this evolving law; evaluating a sentence's plausibility amounts to computing the probability of the corresponding event.

Simple experiments make the same structure visible at smaller scale, with far smaller sample spaces. One die has six elementary outcomes; two dice have thirty-six ordered outcomes; a quality test may have only success and failure; a digital image has an enormous structured state space determined by pixel positions and color intensities. A vocabulary of tens of thousands of tokens, combined over sequences hundreds of tokens long, dwarfs all of these — generative modeling is difficult largely because real sample spaces are huge and their probabilities are far from uniform.

It is worth stressing what softmax does and does not guarantee. It guarantees only the mathematical *form* of a categorical probability law: positivity and normalization. It does not guarantee that the resulting probabilities are well calibrated or correct — whether $P(v_i \mid \text{context})$ tracks the true frequency with which v_i follows that context in language is a statistical property, determined entirely by the training data and loss function, not by the softmax formula itself.

1.4 Combining Outcomes: Addition Rule

1.4.1 Union, intersection, and the complement rule

An event can combine several elementary outcomes. In the two-draws-with-replacement example, the event "same color twice" contains red-red, green-green, and yellow-yellow, so by additivity for mutually exclusive outcomes,

$$P(\text{same color}) = \tfrac{1}{9} + \tfrac{1}{9} + \tfrac{1}{9} = \tfrac{1}{3}.$$

More generally, the union of two events needs a correction whenever they overlap:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B). \tag{1.5}$$

Let A be "at least one red" and B be "two marbles of the same color" in the marble example. Then $P(A) = 5/9$, $P(B) = 3/9$, and $A \cap B$ is red-red with probability $1/9$, so $P(A \cup B) = 7/9$. Simply adding $5/9$ and $3/9$ would double-count red-red. The complement rule follows from the same normalization axiom:

$$P(A^c) = 1 - P(A). \tag{1.6}$$

For disjoint events $A_1, A_2, \ldots$, countable additivity gives

$$P\left(\bigcup_i A_i\right) = \sum_i P(A_i), \qquad \text{when the } A_i \text{ are pairwise disjoint.} \tag{1.7}$$

Splitting $A \cup B$ into three disjoint regions — only A, only B, and the intersection — recovers Equation (1.5); for three or more overlapping events the same idea extends to inclusion–exclusion, which alternates sums and subtractions of intersections.

A *joint-probability table*, with rows for the first draw and columns for the second, is a convenient representation for two categorical variables: each cell is an elementary joint outcome, row and column sums are marginal probabilities, and selected blocks or diagonals represent events. Making overlap visible in a table is a simple, reliable way to avoid double counting.

1.4.2 Team example

Example. A conservative design team, call it C, and an innovative design team, call it N, are asked to separately design a new product within a month. From past experience we know that:

(a) The probability that team C is successful is $2/3$.

(b) The probability that team N is successful is $1/2$.

(c) The probability that at least one team is successful is $3/4$.

If both teams are successful, the design of team N is adopted. Assuming that exactly one successful design is produced, what is the probability that it was designed by team N?

Solution. Let C and N denote the events that team C and team N are successful, respectively. We are given

$$P(C) = \frac{2}{3}, \qquad P(N) = \frac{1}{2}, \qquad P(C \cup N) = \frac{3}{4}.$$

By the addition rule,

$$P(C \cap N) = P(C) + P(N) - P(C \cup N) = \frac{2}{3} + \frac{1}{2} - \frac{3}{4} = \frac{8}{12} + \frac{6}{12} - \frac{9}{12} = \frac{5}{12}.$$

The event that *team N alone* is successful is $N \cap C^c$, with probability

$$P(N \cap C^c) = P(N) - P(C \cap N) = \frac{1}{2} - \frac{5}{12} = \frac{1}{12}.$$

Similarly, the event that *team C alone* is successful is $C \cap N^c$, with probability

$$P(C \cap N^c) = P(C) - P(C \cap N) = \frac{2}{3} - \frac{5}{12} = \frac{3}{12} = \frac{1}{4}.$$

The event that *exactly one* team is successful is the (disjoint) union of these two events, so

$$P(\text{exactly one successful}) = P(C \cap N^c) + P(N \cap C^c) = \frac{1}{4} + \frac{1}{12} = \frac{4}{12} = \frac{1}{3}.$$

Therefore, the probability that the (unique) successful design was produced by team N, given that exactly one team succeeded, is

$$P(N \cap C^c \mid \text{exactly one successful}) = \frac{P(N \cap C^c)}{P(\text{exactly one successful})} = \frac{1/12}{1/3} = \frac{1}{4}.$$

1.5 Conditional Probability

1.5.1 Definition and the changing denominator

Conditional probability restricts attention to cases in which another event is already known to have occurred. The event B becomes the new reference population, so the intersection of A and B must be renormalized by $P(B)$ rather than by the size of the full sample space:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{n(A \cap B)}{n(B)}, \qquad P(B) > 0. \tag{1.8}$$

For a fair die, let $A = \{1, 2, 3\}$ (a result less than four) and $B = \{1, 3, 5\}$ (an odd result). The intersection is $\{1, 3\}$, so

$$P(A \mid B) = \frac{2/6}{3/6} = \frac{2}{3}, \qquad P(A^c \mid B) = \frac{1}{3},$$

and the two conditional probabilities correctly sum to one. This illustrates why the denominator is required: it renormalizes probabilities after the sample space has been restricted to B. Conditioning can also be pictured as filtering and rescaling — first discard outcomes outside B, then keep the portion of B that also belongs to A, and finally divide by the total mass of B.

1.5.2 The mask-and-COVID example

A second illustrative case, used repeatedly below, considers 100 people cross-classified by consistent mask use and whether they contracted COVID (Table 1.2).

Table 1.2: Mask use and COVID outcome (percent of full sample)

	COVID	**No COVID**	**Total**
Mask	15	45	60
No mask	15	25	40
Total	30	70	100

The marginal probabilities are $P(\text{COVID}) = 0.30$ and $P(\text{mask}) = 0.60$. Conditioning on mask use,

$$P(\text{COVID} \mid \text{mask}) = \frac{0.15}{0.60} = 0.25, \qquad P(\text{COVID} \mid \text{mask}^c) = \frac{0.15}{0.40} = 0.375. \tag{1.9}$$

Observing mask use moves the estimated COVID probability from a 30% prior (marginal) probability down to a 25% posterior probability — a 12.5-percentage-point gap between the two conditional rates, and a five-percentage-point drop from the unconditional rate. This is an association visible in the table; whether mask use *caused* the reduction depends on study design, measurement, compliance, and possible confounding, and the table itself is a teaching example rather than evidence for a real-world causal estimate.

1.6 The Chain Rule: Multiplication rule

1.6.1 From conditional probability to the chain rule

The joint event $A \cap B$ occurs when both A and B occur; marginal probabilities refer to each event separately. Rearranging the conditional-probability definition (Equation (1.8)) gives the **chain rule** (equivalently, the multiplication rule), valid whenever the conditioning probability is positive:

$$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A). \tag{1.10}$$

1.6.2 Markov chain example

Example. Alice is taking a probability class and at the end of each week she can be either up-to-date or she may have fallen behind. If she is up-to-date in a given week, the probability that she will be up-to-date (or behind) in the next week is 0.8 (or 0.2, respectively). If she is behind in a given week, the probability that she will be up-to-date (or behind) in the next week is 0.6 (or 0.4, respectively). Alice is (by default) up-to-date when she starts the class. What is the probability that she is up-to-date after three weeks?

Solution. Let U_k denote the event that Alice is up-to-date at the end of week k, and let $B_k = U_k^c$ denote the event that she is behind at the end of week k. We are given the transition probabilities

$$P(U_{k+1} \mid U_k) = 0.8, \qquad P(B_{k+1} \mid U_k) = 0.2,$$

$$P(U_{k+1} \mid B_k) = 0.6, \qquad P(B_{k+1} \mid B_k) = 0.4,$$

and the initial condition $P(U_0) = 1$ (Alice starts up-to-date).

Week 1. By the total probability theorem,

$$P(U_1) = P(U_1 \mid U_0)P(U_0) + P(U_1 \mid B_0)P(B_0) = (0.8)(1) + (0.6)(0) = 0.8,$$

$$P(B_1) = 1 - P(U_1) = 0.2.$$

Week 2. Applying the total probability theorem again,

$$P(U_2) = P(U_2 \mid U_1)P(U_1)+P(U_2 \mid B_1)P(B_1) = (0.8)(0.8)+(0.4)(0.2) = 0.64+0.08 = 0.72,$$

$$P(B_2) = 1 - P(U_2) = 0.28.$$

Week 3. Once more,

$$P(U_3) = P(U_3 \mid U_2)P(U_2) + P(U_3 \mid B_2)P(B_2) = (0.8)(0.72) + (0.4)(0.28) = 0.688.$$

Therefore, the probability that Alice is up-to-date after three weeks is

$$P(U_3) = 0.688.$$

1.7 Independence

Events A and B are **independent** when learning that one occurred does not change the probability of the other. The following statements are equivalent whenever the relevant probabilities are nonzero:

$$A \perp\!\!\!\perp B \iff P(A \mid B) = P(A) \iff P(A \cap B) = P(A)P(B). \tag{1.11}$$

A related diagnostic in binary settings compares $P(A \mid B)$ with $P(A \mid B^c)$: if the two rates differ, B carries information about A. Independence is symmetric — if A is independent of B, then B is independent of A, A is independent of B^c, and A^c is independent of B — and a joint table for independent events factors into the product of its row and column marginals. Independence should not be confused with mutual exclusivity: two mutually exclusive events with positive probability cannot be independent, since the occurrence of one makes the other impossible. Independence is also a property of the joint distribution, not a claim that two events are causally unrelated in every scientific sense.

1.7.1 Covariance of binary event indicators

Let I_A equal 1 when event A occurs and 0 otherwise. For two event indicators, covariance has a direct probability interpretation:

$$\mathrm{Cov}(I_A, I_B) = P(A \cap B) - P(A)P(B). \tag{1.12}$$

Applying this to the mask-and-COVID table (Section 1.5.2),

$$\mathrm{Cov}(I_C, I_M) = 0.15 - (0.30)(0.60) = -0.03.$$

Under independence, the expected joint proportion would be $0.30 \times 0.60 = 0.18$; the observed joint proportion of 0.15 is three percentage points lower, so mask use and COVID occurrence co-occur *less* often than independence would predict in this dataset. A technical caveat is worth stating precisely: zero covariance means *uncorrelated*, not generally *independent*. For two Bernoulli indicators specifically, $\mathrm{Cov}(I_A, I_B) = 0$ is equivalent to $P(A \cap B) = P(A)P(B)$ and so does imply independence of those two binary events; for general random variables, however, nonlinear dependence can persist even when covariance is zero. Refer to the counter example provided in §**??**.

1.7.2 Design matters: the advertisement-and-purchase example

In a stylized group of 100 online consumers, 30 click a displayed advertisement and 12 of those clickers go on to purchase. The critical missing quantity is the number x of purchases among the 70 non-clickers — without that comparison group, no effect or association involving ad engagement can be assessed. The clicker purchase rate is $12/30 = 0.40$; independence requires the same rate among non-clickers, so

$$\frac{x}{70} = \frac{12}{30} = 0.40 \implies x = 28, \tag{1.13}$$

giving 40 total purchasers under the independent benchmark, with 12 among clickers and 28 among non-clickers; equivalently, $P(\text{Purchase} \cap \text{Ad}) = 0.12 = (0.40)(0.30) = P(\text{Purchase})P(\text{Ad})$. The broader design principle: a rate observed only among exposed or engaged users is never enough to establish an effect. A valid comparison needs an appropriate unexposed group and, for a causal claim, a design that addresses selection into exposure — people who click ads may differ systematically from those who do not.

1.7.3 Independence of a collection of events

The definition of independence extends naturally from two events to a whole collection. For three events A_1, A_2, and A_3, independence means satisfying all four of the conditions

$$P(A_1 \cap A_2) = P(A_1)P(A_2), \tag{1.14}$$
$$P(A_1 \cap A_3) = P(A_1)P(A_3), \tag{1.15}$$
$$P(A_2 \cap A_3) = P(A_2)P(A_3), \tag{1.16}$$
$$P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3). \tag{1.17}$$

The first three conditions simply assert that every pair of events is independent, a property called **pairwise independence**. The fourth condition is a separate requirement, and — perhaps surprisingly — it does not follow from the first three: a collection of events can be pairwise independent while still failing the joint condition in Equation (1.17). The converse also fails: satisfying Equation (1.17) alone does not guarantee that the three pairwise conditions in Equations (1.14)–(1.16) hold. The two examples that follow illustrate each direction of this gap, which is why a model that assumes several events are **mutually independent** must state and check all four conditions — pairwise independence checked one pair at a time is not enough.

1.7.4 Pairwise independence does not imply independence

Independence is a genuinely joint property of a collection of events, not something that can be checked one pair at a time. The following example shows three events that are pairwise independent — every pair satisfies the multiplication rule — while the full collection is not independent.

Consider two independent fair coin tosses, and the following events:

$$H_1 = \{\text{1st toss is a head}\}, \qquad H_2 = \{\text{2nd toss is a head}\}$$

and

$$D = \{\text{the two tosses have different results}\}.$$

The events H_1 and H_2 are independent by definition, since the two tosses are independent. To see that H_1 and D are also independent, condition on H_1:

$$P(D \mid H_1) = \frac{P(H_1 \cap D)}{P(H_1)} = \frac{1/4}{1/2} = \frac{1}{2} = P(D).$$

By the same argument, H_2 and D are independent as well. Every pair among H_1, H_2, D therefore satisfies the independence criterion from Equation (1.11).

Yet the three events together are not independent. If H_1 and H_2 both occur, the two tosses agree, so D cannot occur:

$$P(H_1 \cap H_2 \cap D) = 0 \qquad \text{while} \qquad P(H_1)P(H_2)P(D) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

Since $0 \neq 1/8$, the three-way multiplication rule fails, even though it holds for every pair separately. **Pairwise independence, checked two events at a time, is strictly weaker than mutual independence of the whole collection** — a distinction that matters whenever a model assumes several events or random variables are jointly independent rather than merely uncorrelated in pairs.

1.7.5 The triple-intersection condition alone is not enough

The above example showed that pairwise independence does not imply the joint condition $P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3)$. The converse gap is just as real: satisfying that single joint condition does not imply any of the three pairwise conditions.

Consider two independent rolls of a fair die, and the following events:

$$A = \{\text{1st roll is } 1, 2, \text{ or } 3\}, \qquad B = \{\text{1st roll is } 3, 4, \text{ or } 5\}$$

and

$$C = \{\text{the sum of the two rolls is } 9\}.$$

We have

$$\begin{aligned} P(A \cap B) &= \frac{1}{6} \neq \frac{1}{2} \cdot \frac{1}{2} = P(A)P(B), \\ P(A \cap C) &= \frac{1}{36} \neq \frac{1}{2} \cdot \frac{4}{36} = P(A)P(C), \\ P(B \cap C) &= \frac{1}{12} \neq \frac{1}{2} \cdot \frac{4}{36} = P(B)P(C). \end{aligned}$$

Thus the three events A, B, and C are not independent, and indeed *no two* of these events are independent. On the other hand,

$$P(A \cap B \cap C) = \frac{1}{36} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{4}{36} = P(A)P(B)P(C).$$

The triple-intersection condition holds exactly, even though every pairwise condition fails.

The intuition behind the independence of a collection of events is analogous to the two-event case: independence means that the occurrence or non-occurrence of any number of the events in the collection carries no information about the remaining events or their complements. For example, if the events A_1, A_2, A_3, A_4 are independent, one obtains relations such as

$$P(A_1 \cup A_2 \mid A_3 \cap A_4) = P(A_1 \cup A_2)$$

or

$$P(A_1 \cup A_2^c \mid A_3^c \cap A_4) = P(A_1 \cup A_2^c).$$

1.8 Conditional independence

Two events that are marginally (pairwise) dependent may become independent once a third event is conditioned on:

Definition 1 (Conditional independence)**.** Two events A and B are said to be *conditionally independent*, given another event C with $P(C) > 0$, if

$$A \perp B \mid C \quad \Longleftrightarrow \quad P(A \cap B \mid C) = P(A \mid C)\, P(B \mid C).$$

Theorem 1. *If, in addition, $P(B \cap C) > 0$, conditional independence is equivalent to the condition*

$$A \perp B \mid C \quad \Longleftrightarrow \quad P(A \mid B \cap C) = P(A \mid C). \tag{1.18}$$

Proof. The definition of the conditional probability and the multiplication rule yield

$$P(A \cap B \mid C) = \frac{P(A \cap B \cap C)}{P(C)} = \frac{P(C)\,P(B \mid C)\,P(A \mid B \cap C)}{P(C)} = P(B \mid C)\,P(A \mid B \cap C).$$

Recall that by definition of conditional independence,

$$P(A \cap B \mid C) = P(A \mid C)\,P(B \mid C).$$

Since $P(B \cap C) > 0$, we have $P(B \mid C) > 0$, so we may cancel this factor to obtain that conditional independence, $P(A \cap B \mid C) = P(A \mid C)\,P(B \mid C)$, holds if and only if

$$P(A \mid B \cap C) = P(A \mid C),$$

which is precisely (1.18). □

In words, this relation states that if C is known to have occurred, the additional knowledge that B also occurred does not change the probability of A.

Conditional independence is central to larger probabilistic models: graphical models, naive Bayes, hidden Markov models, and many sequence models all exploit conditional-independence structure to avoid estimating an impossibly large, unrestricted joint distribution over all variables at once.

Interestingly, independence of two events A and B with respect to the unconditional probability law, does not imply conditional independence, and vice versa, as illustrated by the next two examples.

Example (Two biased coins)**.** There are two coins, a blue and a red one. We choose one of the two at random, each being chosen with probability 1/2, and proceed with two independent tosses. The coins are biased: with the blue coin, the probability of heads in any given toss is 0.99, whereas for the red coin it is 0.01.

Let B be the event that the blue coin was selected. Let also H_i be the event that the ith toss resulted in heads. Given the choice of a coin, the events H_1 and H_2 are independent, because of our assumption of independent tosses. Thus,

$$P(H_1 \cap H_2 \mid B) = P(H_1 \mid B)\,P(H_2 \mid B) = 0.99 \cdot 0.99.$$

On the other hand, the events H_1 and H_2 are *not* independent. Intuitively, if we are told that the first toss resulted in heads, this leads us to suspect that the blue coin was selected, in which case, we expect the second toss to also result in heads.

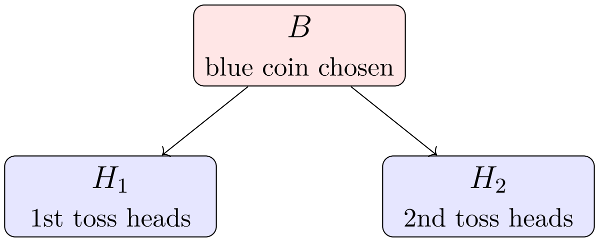
Mathematically, we use the total probability theorem to obtain

$$P(H_1) = P(B)P(H_1 \mid B) + P(B^c)P(H_1 \mid B^c) = \frac{1}{2} \cdot 0.99 + \frac{1}{2} \cdot 0.01 = \frac{1}{2},$$

as should be expected from symmetry considerations. Similarly, we have $P(H_2) = 1/2$. Now notice that

$$P(H_1 \cap H_2) = P(B)P(H_1 \cap H_2 \mid B) + P(B^c)P(H_1 \cap H_2 \mid B^c) = \frac{1}{2} \cdot 0.99 \cdot 0.99 + \frac{1}{2} \cdot 0.01 \cdot 0.01 \approx \frac{1}{2}.$$

Thus, $P(H_1 \cap H_2) \neq P(H_1)P(H_2)$, and the events H_1 and H_2 are dependent, even though they are conditionally independent given B.



No arrow between H_1 and H_2 – dependent only through their shared cause B

Figure 1.2: Fork (common-cause) structure for Example.

Remark (Graphical intuition). Biased coin example can be understood through the following graphical structure, sometimes called a *fork*:

$$H_1 \longleftarrow B \longrightarrow H_2.$$

Both H_1 and H_2 are caused by (depend on) which coin was selected, B, but there is no arrow directly connecting H_1 and H_2 themselves.

Given B (a specific coin is fixed), the arrow from B into H_1 and H_2 has already "done its job": once we know which coin is being used, the two tosses are simply independent flips of that one coin. There is no remaining path connecting H_1 and H_2, so $P(H_1 \cap H_2 \mid B) = P(H_1 \mid B)\, P(H_2 \mid B)$. This is why H_1 and H_2 are *conditionally independent given* B.

Unconditionally (marginalizing over B), H_1 and H_2 become statistically linked *through* B – the fork is "open." Observing $H_1 =$ heads makes it more likely that the blue coin was chosen, which in turn makes $H_2 =$ heads more likely. This is why $P(H_1 \cap H_2) \neq P(H_1)\, P(H_2)$, despite H_1 and H_2 being conditionally independent given B.

This is the mirror image of the collider structure $H_1 \to D \leftarrow H_2$ from coin toss example below: there, conditioning on D *created* dependence between two unconditionally independent causes; here, conditioning on B *removes* dependence that exists unconditionally because B is a shared cause. Recognizing which structure – fork or collider – a set of variables forms tells us immediately whether conditioning will create or destroy dependence between them.

Example (Conditional dependence despite unconditional independence)**.** Consider two independent fair coin tosses, in which all four possible outcomes are equally likely. Let

$$H_1 = \{\text{1st toss is a head}\}, \qquad H_2 = \{\text{2nd toss is a head}\}$$

and

$$D = \{\text{the two tosses have different results}\}.$$

The events H_1 and H_2 are (unconditionally) independent. But

$$P(H_1 \mid D) = \frac{1}{2}, \qquad P(H_2 \mid D) = \frac{1}{2}, \qquad P(H_1 \cap H_2 \mid D) = 0,$$

so that $P(H_1 \cap H_2 \mid D) \neq P(H_1 \mid D)\, P(H_2 \mid D)$, and H_1, H_2 are not conditionally independent.

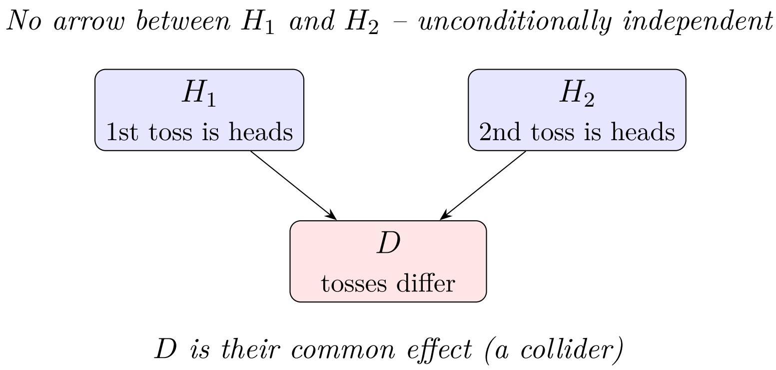


Figure 1.3: Collider structure for Example

Remark (Graphical intuition). Coin toss example can be understood through the following graphical structure, sometimes called a *collider*:

$$H_1 \longrightarrow D \longleftarrow H_2.$$

Both H_1 and H_2 point into D, but there is no arrow directly connecting H_1 and H_2 – reflecting the fact that they are *unconditionally* independent.

Unconditionally, knowing H_1 tells us nothing about H_2: no path connects them except through D, and until we condition on D, that path is "closed." This is why $P(H_1 \cap H_2) = P(H_1)\,P(H_2)$.

Once we condition on D (i.e., we are told whether the tosses differed), we effectively "open" the path between H_1 and H_2 through the collider. Information can now flow between H_1 and H_2: if we know D occurred (the tosses differed) and we learn H_1 (the first toss was heads), we can immediately deduce H_2^c (the second toss must have been tails). This is exactly why

$$P(H_1 \cap H_2 \mid D) = 0 \neq \frac{1}{4} = P(H_1 \mid D)\,P(H_2 \mid D).$$

This phenomenon is known as *explaining away*: two independent causes of a shared, observed effect become entangled once we know the effect occurred, because learning about one cause changes what we can infer about the other. A familiar example: if a car will not start (D) and we learn the battery is fine (H_1), this makes it *more* likely that the starter is broken (H_2), even though battery health and starter health are unrelated in general.

Remark (The three basic graphical configurations). For a triple of variables connected by two edges – the building block of every larger graphical model – there are exactly three basic configurations.

Chain (mediator): $A \to B \to C$. Here A influences C only *through* B: B is an intermediate step on a causal path running from A to C. Unconditionally, A and C are dependent,

$$P(A \cap C) \neq P(A)\,P(C),$$

but once we condition on B, the path is blocked, and

$$P(A \cap C \mid B) = P(A \mid B)\, P(C \mid B), \qquad \text{i.e.} \quad A \perp C \mid B.$$

Fork (common cause): $A \leftarrow B \rightarrow C$. Here B is not an intermediate step but a shared *root* cause that generates both A and C independently. Despite this structural difference from the chain, the conditional independence pattern is identical:

$$P(A \cap C) \neq P(A)\, P(C), \qquad \text{but} \qquad P(A \cap C \mid B) = P(A \mid B)\, P(C \mid B), \qquad \text{i.e.} \quad A \perp C \mid B.$$

Chains and forks are statistically indistinguishable. This is a crucial and somewhat surprising point: from probabilities and conditional independence relations *alone*, there is no way to tell whether $A \rightarrow B \rightarrow C$ or $A \leftarrow B \rightarrow C$ is the true structure – both produce exactly the same pattern, $A \not\perp C$ unconditionally and $A \perp C \mid B$ given B. The two are said to belong to the same *Markov equivalence class*. Distinguishing a chain from a fork requires information outside the joint distribution itself – for instance, knowledge of temporal order (causes precede effects) or the result of an intervention (e.g., manipulating B directly and observing whether A changes).

Collider (common effect): $A \rightarrow B \leftarrow C$. Here B is a shared effect of A and C. This case is the one that *is* statistically identifiable, because its pattern is the opposite of the chain and the fork:

$$P(A \cap C) = P(A)\, P(C), \qquad \text{i.e.} \quad A \perp C,$$

unconditionally, but

$$P(A \cap C \mid B) \neq P(A \mid B)\, P(C \mid B),$$

so A and C are *not* conditionally independent given B. Conditioning on B *opens* a path between A and C and makes them dependent, even though they are independent unconditionally.

The key pattern to remember is that chains and forks behave alike – conditioning on the middle node *blocks* dependence, turning $A \not\perp C$ into $A \perp C \mid B$, and are indistinguishable from data alone – while colliders behave oppositely – conditioning on the middle node *creates* dependence, turning $A \perp C$ into $A \not\perp C \mid B$, and are the one configuration identifiable purely from conditional independence relations. This asymmetry is the reason colliders are the "surprising" case, and it is also what makes them so useful in causal discovery algorithms: finding a collider pattern in data is direct evidence of the underlying causal structure, whereas finding a chain/fork pattern is not.

There is no fourth basic pattern for a three-node, two-edge subgraph; any larger graph is built by composing these three motifs repeatedly. This is the basis of the *d-separation* criterion in Bayesian networks, which determines conditional independence in arbitrarily large graphs by checking, for every path between two nodes, whether it passes through a blocked chain/fork or an unblocked collider.

1.9 Case study: Simpson's Paradox

1.9.1 Applying conditioning: acceptance by gender

Returning to the admissions data, the raw acceptance rates by gender are shown in Table 1.3.

Table 1.3: Aggregate acceptance rate by gender

Group	**Applicants** $n(g)$	**Accepted** $n(A \cap g)$	**Acceptance** $P(A \mid g)$
Men	1,963	967	$967/1{,}963 = 49.3\%$
Women	881	339	$339/881 = 38.5\%$
All applicants	2,844	1,306	$1{,}306/2{,}844 = 45.9\%$

The raw difference is $38.5\% - 49.3\% = -10.8$ percentage points for women relative to men. At this stage the gap is purely descriptive: it does not by itself establish discrimination, statistical significance, or causality, since department choice may be associated with both gender and department selectivity — the subject of the next section.

1.9.2 Department-specific results

A finer, post-hoc analysis conditions jointly on gender and department, examining $P(A \mid g, D = d)$ instead of $P(A \mid g)$ alone. Table 1.4 reports acceptance rates within each department.

Table 1.4: Acceptance rates within department

Department	**Men acc. / n**	**Men rate**	**Women acc. / n**	**Women rate**	**W−M**
Management	496 / 616	80.5%	155 / 175	88.6%	+8.1 pp
Law	53 / 272	19.5%	94 / 481	19.5%	+0.1 pp
Engineering	418 / 1,075	38.9%	90 / 225	40.0%	+1.1 pp
Aggregate	967 / 1,963	49.3%	339 / 881	38.5%	−10.8 pp

Within every single department, women's acceptance rate is equal to or higher than men's; yet the aggregate comparison reverses to favor men by 10.8 percentage points. This reversal is a textbook instance of **Simpson's paradox**: an association that holds in every subgroup disappears, or even flips, once the subgroups are pooled. Figure 1.4 illustrates the reversal.

1.9.3 The mechanism: different weights, not different truths

An aggregate conditional rate is a weighted average of department-specific rates, with weights given by how each gender's applications are distributed across departments:

$$P(A \mid g) = \sum_{d \in \mathcal{D}} P(A \mid g, D = d)\, P(D = d \mid g). \tag{1.19}$$

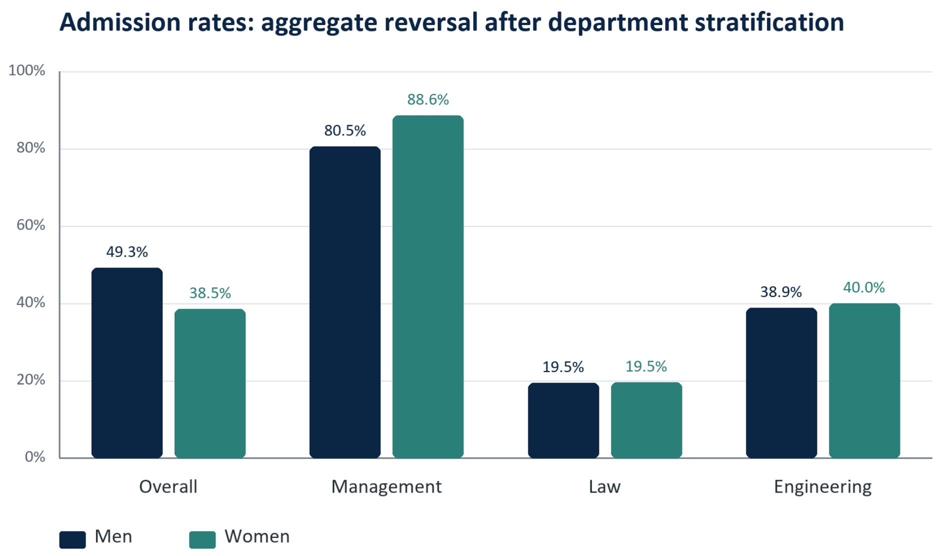


Figure 1.4: The aggregate gender comparison reverses after conditioning on department.

Table 1.5 shows the mix driving the reversal: men are far more concentrated in Management, the *least* selective department, while women are far more concentrated in Law, the *most* selective department.

Table 1.5: Application mix by gender and department selectivity

Department	$P(D = d \mid \textbf{Men})$	$P(D = d \mid \textbf{Women})$	**Selectivity** $P(A \mid D = d)$
Management	$616/1{,}963 = 31.4\%$	$175/881 = 19.9\%$	$651/791 = 82.3\%$
Law	$272/1{,}963 = 13.9\%$	$481/881 = 54.6\%$	$147/753 = 19.5\%$
Engineering	$1{,}075/1{,}963 = 54.8\%$	$225/881 = 25.5\%$	$508/1{,}300 = 39.1\%$

Applying Equation (1.19) explicitly,

$$P(A \mid M) = 0.805(0.314) + 0.195(0.139) + 0.389(0.548) = 0.493, \tag{1.20}$$
$$P(A \mid F) = 0.886(0.199) + 0.195(0.546) + 0.400(0.255) = 0.385. \tag{1.21}$$

Nearly equal within-department rates, combined with very different weights, reproduce the aggregate rates from Table 1.3. Simpson's paradox is not a mathematical contradiction: the aggregate and stratified quantities condition on different information and apply different weights, so a responsible analysis states exactly which estimand is being interpreted and why it matches the research question.

One descriptive way to remove the compositional difference is *standardization*: apply a common department mix — here the overall proportions of 27.8% Management, 26.5% Law, and 45.7% Engineering — to both genders. This produces standardized rates of 45.3% for men and 48.1% for women, a swing to +2.8 percentage points in favor of women. Standardization is an adjusted descriptive comparison, not by itself a causal estimate.

1.9.4 Simpson's paradox as a mediator pattern

Remark (Simpson's paradox as a mediator pattern). The reversal in Section 1.9 can be understood as an instance of the *chain (mediator)* structure introduced earlier,

$$G \longrightarrow D \longrightarrow A,$$

illustrated in Figure 1.5. Gender G influences which department a candidate applies to, and department D in turn influences the probability of acceptance A; there is no arrow directly from G to A other than through this mediating path.

Unconditionally (i.e., pooling across departments), G and A are dependent,

$$P(A \mid G) \neq P(A),$$

and in this data set the pooled comparison, $P(A \mid \text{Men}) = 49.3\%$ versus $P(A \mid \text{Women}) = 38.5\%$, appears to favor men.

Once we condition on the mediator D, however, the picture changes completely: within every department, the association between G and A is negligible or reversed (Table 1.4), consistent with

$$P(A \mid G, D = d) \approx P(A \mid D = d) \qquad \text{for each department } d,$$

i.e., G and A are approximately conditionally independent given D. This is exactly the behavior predicted by the chain structure: the marginal dependence between G and A seen in the aggregate table is not evidence of a direct effect of gender on acceptance, but is instead *entirely explained by* the indirect path through department, combined with the fact that men and women apply to departments in very different proportions (Table 1.5).

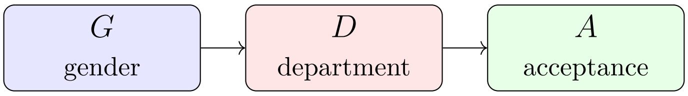
This is precisely why Equation (1.19) is the right tool for diagnosing the paradox: it decomposes the marginal (unconditioned) quantity $P(A \mid G)$ into a weighted average of the conditional (mediator-adjusted) quantities $P(A \mid G, D = d)$, with weights $P(D = d \mid G)$ that differ sharply by gender. The paradox arises precisely because pooling over D mixes together department-specific rates using very different weights for men and women, producing an aggregate comparison that need not resemble any of the department-specific comparisons.

As with the chain and fork structures generally, this mediator pattern alone cannot establish that $G \to D \to A$, rather than some other explanation (e.g., an unobserved confounder affecting both department choice and acceptance standards), is the correct causal story – that requires substantive knowledge of the admissions process, not the conditional independence pattern alone. What the data *can* establish is the descriptive fact that the department-level comparisons and the aggregate comparison answer different questions, and that conflating them is the source of the apparent paradox.

1.10 Case study: Berkson's Paradox and Selection Bias

1.10.1 Setting up the data

Consider an applicant pool in which candidates are evaluated on two attributes: a standardized *test score*, coded T = high or T = low, and an *interview score*, coded



G influences A only through D – a chain (mediator) structure

Figure 1.5: Mediator structure underlying Simpson's paradox: gender influences department choice, and department influences acceptance.

I = high or I = low. Among 800 applicants, the two attributes are, by construction, statistically independent and evenly split, as shown in Table 1.6.

Table 1.6: Joint distribution of test and interview outcomes (all applicants)

	Interview High	**Interview Low**	**Total**
Test High	200	200	400
Test Low	200	200	400
Total	400	400	800

Directly from the table,

$$P(T = \text{high}) = \frac{400}{800} = 50\%, \qquad P(I = \text{high}) = \frac{400}{800} = 50\%$$

and

$$P(T = \text{high} \cap I = \text{high}) = \frac{200}{800} = 25\%,$$

and since $25\% = 50\% \times 50\%$, test performance and interview performance are **unconditionally independent**: $T \perp I$.

1.10.2 Applying an admission rule (introducing the collider)

Suppose the admissions committee admits any candidate who performs well on *at least one* dimension, and rejects only candidates who are weak on both:

$$A = 1 \text{ (admitted)} \iff T = \text{high or } I = \text{high}.$$

This rule makes admission A a common effect of T and I – exactly the collider structure introduced earlier. Table 1.7 shows how each cell of the original population is resolved under this rule.

Table 1.7: Admission outcome by test/interview combination

Combination	**Applicants**	**Outcome**	**Admitted?**
Test High, Interview High	200	Strong on both	Yes
Test High, Interview Low	200	Strong on test only	Yes

Table 1.7: Admission outcome by test/interview combination

Combination	**Applicants**	**Outcome**	**Admitted?**
Test Low, Interview High	200	Strong on interview only	Yes
Test Low, Interview Low	200	Weak on both	No
Total admitted	600		
Total rejected	200		

1.10.3 What happens when we condition on the collider

Table 1.8 restricts attention to the 600 *admitted* candidates only – i.e., it shows the joint distribution of T and I conditional on $A = 1$.

Table 1.8: Joint distribution of test and interview outcomes, conditional on admission

	Interview High	**Interview Low**	**Total**
Test High	200	200	400
Test Low	200	0	200
Total	400	200	600

Now compute the conditional marginals and the conditional joint probability:

$$P(T = \text{high} \mid A = 1) = \frac{400}{600} \approx 66.7\%, \qquad P(I = \text{high} \mid A = 1) = \frac{400}{600} \approx 66.7\%,$$

$$P(T = \text{high} \cap I = \text{high} \mid A = 1) = \frac{200}{600} \approx 33.3\%.$$

If T and I were still independent given $A = 1$, we would expect $P(T = \text{high} \mid A = 1) \cdot P(I = \text{high} \mid A = 1) \approx 0.667 \times 0.667 \approx 44.4\%$. The actual conditional joint probability, 33.3%, is noticeably lower, so

$$P(T = \text{high} \cap I = \text{high} \mid A = 1) \neq P(T = \text{high} \mid A = 1)\, P(I = \text{high} \mid A = 1),$$

and T, I are **no longer independent once we condition on admission**.

The effect is even starker if we look at one specific comparison:

$$P(I = \text{high} \mid T = \text{high},\, A = 1) = \frac{200}{400} = 50\%, \qquad P(I = \text{high} \mid T = \text{low},\, A = 1) = \frac{200}{200} = 100\%.$$

Among admitted candidates, a *weak* test score makes a strong interview *certain*, while a strong test score leaves the interview outcome a coin flip. Test performance and interview performance, entirely unrelated in the applicant pool, appear strongly *negatively* associated once we look only at who was admitted.

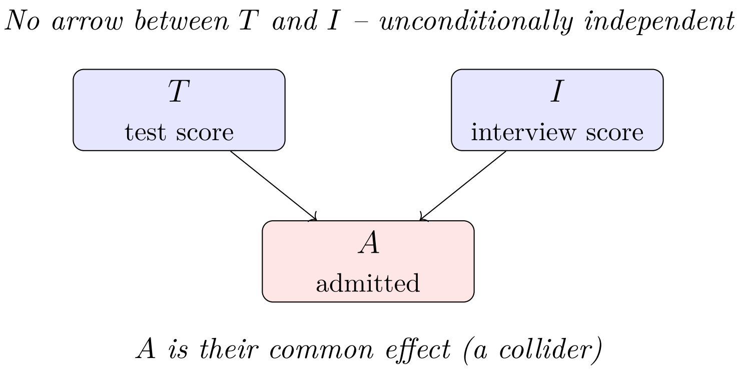


Figure 1.6: Collider structure underlying the admissions selection effect: test score and interview score are independent causes of admission.

1.10.4 The mechanism: conditioning on a collider

Remark (Berkson's paradox). The pattern in this case study is a classical instance of the *collider* structure,

$$T \longrightarrow A \longleftarrow I,$$

illustrated in Figure 1.6, and is often called *Berkson's paradox* or, more generally, *selection bias*. Both T and I influence A, but there is no direct arrow between T and I themselves – consistent with their unconditional independence.

Restricting the data to admitted candidates is precisely conditioning on the collider A. As with the biased-coins collider example earlier in these notes, this opens a path between T and I: knowing that a candidate was admitted *and* that their test score was weak is enough to infer, with high confidence, that their interview score was strong – one explanation for admission "explains away" the need for the other. This is exactly the same phenomenon as before, just with D ("tosses differ") replaced by A ("admitted").

Berkson's paradox is a common source of misleading correlations in observed (rather than randomly sampled) data, because the population under study has already been filtered by some selection process related to the variables of interest:

- Studies of hospitalized patients can find spurious negative associations between two unrelated diseases, because having *either* disease increases the chance of being in the hospital sample in the first place.
- Observational studies of successful entrepreneurs or celebrities can find spurious trade-offs between traits (e.g., talent and luck), because reaching that select group typically requires being strong on at least one dimension.
- In this admissions example, a naive analyst comparing test scores and interview scores *only among admitted students* could mistakenly conclude that the two are competing or substitutable signals, when in the full applicant pool they are completely unrelated.

As in the mediator case study of Simpson's paradox, the lesson is methodological: an association computed on a selected subsample answers a different question than the same association computed on the full population, and conflating the two can manufacture relationships – or, as in Simpson's paradox, reverse them – that do not exist in the underlying data-generating process.

1.10.5 Colliders and "bad controls" in regression

Remark (Colliders and "bad controls" in regression). The collider structure has an important practical consequence for applied regression analysis. Suppose we wish to estimate the effect of test score T and interview score I on a downstream outcome such as salary, using a model of the form

$$\text{Salary} = \beta_0 + \beta_1 T + \beta_2 I + \varepsilon.$$

A natural instinct is to also control for admission status A, since A is observed and clearly related to both regressors. This instinct should be resisted.

Why A should not be added as a control. As established in the case study above, A is a *collider* of T and I,

$$T \longrightarrow A \longleftarrow I.$$

Conditioning on a collider opens a spurious statistical path between its causes. Adding A as a regressor conditions on it in exactly this sense: even if T and I each have a clean, independent causal effect on salary, their estimated coefficients $\hat{\beta}_1, \hat{\beta}_2$ can become biased – inflated, deflated, or even reversed in sign – once A is included, because A is a *descendant of the regressors* rather than a common cause of the regressors and the outcome. In the language of causal inference, A is what Angrist and Pischke [2009] call a **bad control**: a post-treatment variable that should be excluded, not included, from the regression.

This should be contrasted with the usual advice to control for confounders. If A were instead a *fork* – a common cause of T, I, and salary – omitting it would bias the estimated effects of T and I through the usual omitted-variable-bias mechanism, and it *should* be included. Whether a variable belongs in the regression therefore depends critically on its graphical role (confounder vs. collider/descendant), not merely on whether it is correlated with the regressors.

A second, subtler problem: sample selection. Even if A is correctly excluded from the regressors, the bias can re-enter through the data itself. Salaries are typically observable only for *admitted* candidates – rejected applicants generate no salary outcome to record. In that case, the estimation sample is already restricted to $A = 1$ before any model is specified, so the regression is implicitly conditioned on the collider regardless of which variables appear on the right-hand side. This is the classical **sample selection bias** problem [Heckman, 1979], and dropping A from the regressor list does *not* fix it, because the selection occurred at the data-collection stage, not the model-specification stage. Addressing it requires a different tool, such as a Heckman two-step selection correction or an instrument for admission, rather than a simple choice of controls.

Takeaway. When deciding whether to include a variable in a regression, it is not enough to ask whether it is correlated with the outcome or the regressors of interest. One must ask what graphical role the variable plays: a confounder (fork) should typically be controlled for, while a collider – or any variable downstream of the regressors, including one that implicitly defines which observations appear in the sample at all – should not.

1.11 Bayesian approach: Bayes' Rule and Probabilistic Updating

1.11.1 From the chain rule to Bayes' rule

Equating the two right-hand sides of the chain rule (Equation (1.10)) and solving for $P(A \mid B)$ produces **Bayes' rule**:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}. \tag{1.22}$$

Bayes' rule carries two readings at once: it reverses the direction of conditioning, deriving $P(A \mid B)$ from $P(B \mid A)$; and it updates a *prior* belief $P(A)$ into a *posterior* belief $P(A \mid B)$ using the *likelihood* $P(B \mid A)$ and the total *evidence* $P(B)$. The denominator ensures posterior probabilities across competing hypotheses still sum to one. When A is a hypothesis and its complement A^c is the only alternative, the denominator expands by the **law of total probability**:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B \mid A)P(A) + P(B \mid A^c)P(A^c)}. \tag{1.23}$$

More generally, if $B_1, \ldots, B_k$ form a mutually exclusive and exhaustive partition of Ω, the law of total probability reconstructs a marginal probability from conditional pieces,

$$P(A) = \sum_{i=1}^{k} P(A \mid B_i)P(B_i), \tag{1.24}$$

and the same sum, computed over all hypotheses, replaces the two-term denominator of Equation (1.23) for a partition with more than two pieces. In a diagnostic setting A might be a positive test and the B_i disease states; in a forecasting setting A might be a positive signal and the partition good versus bad economic conditions.

Returning to the mask example, $P(\text{mask} \mid \text{COVID}) = 0.5$, since half of the COVID group wears masks. Bayes' rule recovers the same posterior found directly from the table in Equation (1.9):

$$P(\text{COVID} \mid \text{mask}) = \frac{0.50 \times 0.30}{0.60} = 0.25.$$

1.11.2 Base rates and posterior odds

Bayes' rule is most informative — and most often misapplied — when base rates matter. A high $P(B \mid A)$ does not automatically imply a high $P(A \mid B)$; the posterior combines the likelihood with the prior prevalence and divides by the overall evidence, and ignoring $P(A)$ is the classic *base-rate fallacy*. An equivalent technical form separates prior belief from evidential strength using odds:

$$\text{Posterior odds of } A \text{ vs. } A^c = \text{prior odds} \times \text{likelihood ratio}, \qquad \text{likelihood ratio} = \frac{P(B \mid A)}{P(B \mid A^c)}. \tag{1.25}$$

A likelihood ratio above one favors A; below one favors A^c; exactly one carries no information about A at all. In the mask table, this comparison is purely descriptive: conditioning identifies an association, not a cause, and a causal conclusion would require stronger design assumptions about confounders, selection, and measurement.

1.11.3 Worked example: the supplier-quality problem

A company receives 60% of its parts from Montreal (M) and 40% from Toronto (T); the good-part rates are 96% for Montreal and 98% for Toronto. A part is observed to be good (G), but its supplier label is missing, and the question is $P(T \mid G)$.

Table 1.9: Prior supplier shares and quality likelihoods

Supplier	**Prior** $P(\textbf{Supplier})$	**Good rate** $P(G \mid \textbf{Supplier})$	**Joint** $P(G \cap \textbf{Supplier})$
Montreal (M)	0.60	0.96	$0.60 \times 0.96 = 0.576$
Toronto (T)	0.40	0.98	$0.40 \times 0.98 = 0.392$
Total good	1.00	—	$0.576 + 0.392 = 0.968$

$$P(T \mid G) = \frac{(0.40)(0.98)}{(0.40)(0.98) + (0.60)(0.96)} = \frac{0.392}{0.968} = 0.405. \tag{1.26}$$

The prior probability of Toronto was 0.400; observing a good part raises it only slightly, to 0.405, because Toronto's good-part rate is just barely higher than Montreal's. Evidence changes a prior substantially only when it discriminates strongly among the competing hypotheses — here the likelihoods 0.98 and 0.96 are close, so the update is small. As an instructive contrast, observing a *defective* part moves the probability the other way: $P(T \mid \text{defective}) = (0.40 \times 0.02)/[(0.40 \times 0.02) + (0.60 \times 0.04)] = 0.25$, since a defect is relatively more compatible with Montreal, whose defect rate is twice Toronto's.

Table 1.10 organizes the four moving pieces of any Bayesian update using this example.

Table 1.10: Components of a Bayesian update

Component	**Notation**	**Meaning in the supplier example**
Prior	$P(T)$	Supplier probability before seeing the part: 0.40
Likelihood	$P(G \mid T)$	Probability of a good part if from Toronto: 0.98
Evidence	$P(G)$	Overall probability of a good part: 0.968
Posterior	$P(T \mid G)$	Updated supplier probability after seeing a good part: 0.405

The same logic underlies adaptive prediction systems: as a user supplies new information, a model updates its probabilities over possible next actions or outcomes. Sequential learning from evidence is the general principle, though real production systems may combine Bayesian updating with other, non-Bayesian algorithms.

1.11.4 Radar detection example

Example (Radar detection)**.** If an aircraft is present in a certain area, a radar correctly registers its presence with probability 0.99. If it is not present, the radar falsely registers an aircraft presence with probability 0.10. We assume that an aircraft is present with probability 0.05. What is the probability of false alarm (a false indication of aircraft presence), and the probability of missed detection (nothing registers, even though an aircraft is present)? What is $P(\text{aircraft present} \mid \text{radar registers})$?

Solution. Let A denote the event that an aircraft is present, and let R denote the event that the radar registers (indicates) a presence. We are given

$$P(A) = 0.05, \qquad P(A^c) = 0.95,$$

$$P(R \mid A) = 0.99, \qquad P(R^c \mid A) = 0.01,$$
$$P(R \mid A^c) = 0.10, \qquad P(R^c \mid A^c) = 0.90.$$

Probability of false alarm. This is the probability that the radar registers even though no aircraft is present,

$$P(\text{false alarm}) = P(R \cap A^c) = P(R \mid A^c)\,P(A^c) = (0.10)(0.95) = 0.095.$$

Probability of missed detection. This is the probability that the radar fails to register even though an aircraft is present,

$$P(\text{missed detection}) = P(R^c \cap A) = P(R^c \mid A)\,P(A) = (0.01)(0.05) = 0.0005.$$

Probability aircraft is present given the radar registers. By the total probability theorem,

$$P(R) = P(R \mid A)P(A) + P(R \mid A^c)P(A^c) = (0.99)(0.05) + (0.10)(0.95) = 0.0495 + 0.095 = 0.1445.$$

By Bayes' rule,

$$P(A \mid R) = \frac{P(R \mid A)\,P(A)}{P(R)} = \frac{0.0495}{0.1445} \approx 0.3425.$$

Thus, even after the radar registers a detection, there is only about a 34.25% chance that an aircraft is actually present — a consequence of the aircraft being fairly rare ($P(A) = 0.05$) combined with a non-negligible false alarm rate.

1.12 Association, Significance, and Causality

Every example in this lecture — the admissions gap, the mask-and-COVID table, the ad-and-purchase design, the supplier data — raises three distinct questions that require different kinds of evidence to answer, summarized in Table 1.11.

Table 1.11: Three levels of empirical interpretation

Question	**Typical quantity**	**What it establishes**	**What it does not establish**
Descriptive association	Rate difference, covariance, correlation	Variables move together in the observed data	Population stability or causal direction

Table 1.11: Three levels of empirical interpretation

Question	**Typical quantity**	**What it establishes**	**What it does not establish**
Statistical significance	Standard error, confidence interval, p-value	Observed association is difficult to explain by a specified sampling model under H_0	Practical importance or causality
Causal effect	Randomized contrast or identified causal estimand	Outcome change attributable to an intervention under assumptions	Universal validity beyond the design and target population

A nonzero correlation or covariance can arise because A affects B, B affects A, a third variable affects both, the sample is selected, variables are measured imperfectly, or the association is simply due to chance. Causal claims always require research design and assumptions beyond a nonzero covariance. This caution extends to research practice itself: when an observational dataset suggests an interesting pattern but a costly follow-up experiment fails to reproduce it, the second result must not be discarded merely because it is inconvenient. Observational evidence may be confounded, and randomized evidence may reveal that an initial relationship was never causal; a credible research report documents the full sequence of evidence, its limitations, and the differences in design.

1.13 Sequential Probability and Generative AI

The tools of this lecture — conditioning, the chain rule, and conditional independence — are exactly the machinery behind sequence generation in modern AI. For language, the chain rule decomposes a sentence's probability into a product of next-token conditionals:

$$P(w_1, \ldots, w_T) = \prod_{t=1}^{T} P(w_t \mid w_1, \ldots, w_{t-1}). \tag{1.27}$$

A language model estimates each of these factors, normalized into a valid distribution by softmax (Equation (1.4)); generating text then means sampling or selecting from each next-token distribution and appending the chosen token to the conditioning context for the next step. Just as conditional independence lets graphical models and hidden Markov models avoid an intractable full joint distribution (Section 1.6), a language model's contextual conditioning lets it avoid separately estimating the probability of every possible sentence. The probability module of this course therefore connects directly to the computational behavior of the generative systems built on top of it.

1.14 Compact Formula Reference

Table 1.12: Core probability formulas

Concept	**Formula**	**Interpretation**
Empirical probability	$\widehat{P}(E) = n(E)/N$	Fraction of observed units satisfying E
Complement	$P(A^c) = 1 - P(A)$	Probability that A does not occur
Union	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	Correct for double-counted overlap
Conditional probability	$P(A \mid B) = P(A \cap B)/P(B)$	Probability of A within subgroup B
Chain rule	$P(A \cap B) = P(A \mid B)P(B)$	Build a joint probability in stages
Total probability	$P(B) = \sum_i P(B \mid H_i)P(H_i)$	Add mutually exclusive routes to B
Independence	$P(A \cap B) = P(A)P(B)$	The joint equals the product of marginals
Binary covariance	$\mathrm{Cov}(I_A, I_B) = P(A \cap B) - P(A)P(B)$	Observed joint minus independent benchmark
Bayes' rule	$P(A \mid B) = P(B \mid A)P(A)/P(B)$	Reverse conditioning and update a prior
Aggregation	$P(A \mid g) = \sum_d P(A \mid g, d)P(d \mid g)$	Overall rate is a weighted subgroup average
Softmax	$\mathrm{softmax}(z_i) = e^{z_i} / \sum_j e^{z_j}$	Normalize scores into a categorical distribution

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